

Junsu Kim

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Research Interests

Computer Architecture, Systems for ML & ML for Systems

Education

The University of British Columbia, Vancouver, BC, Canada

Sep. 2025 - Current

Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Prashant Nair)

Cumulative GPA: 4.0/4.0

Korea University, Seoul, Korea

Sep. 2023 - Aug. 2025

M.S. in Electrical Engineering (Advisor: Prof. Yunho Oh)

Cumulative GPA: 4.0/4.0

Hanyang University, Seoul, Korea

Mar. 2014 - Feb. 2021

B.S. in Electronic Engineering (Advisor: Prof. Ki-Seok Chung)

Cumulative GPA: 3.81/4.0 (Summa Cum Laude)

Publications

Conference [C] and Journal [J] Papers (* denotes co-first author)

[C6] Jeonghyun Woo*, **Junsu Kim***, Aamer Jaleel, and Prashant Nair, “Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense”, The 53rd International Symposium on Computer Architecture (ISCA), 2026.

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, and Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh “Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture” The IEEE Computer Architecture Letters (CAL), 2025.

[C5] **Junsu Kim**, and Suhyun Kim, “Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning”, The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[C4] Seondeok Kim*, Sangun Choi*, Jaebeom Jeon, **Junsu Kim**, Minseong Gil, Jaehyeok Ryu, and Yunho Oh, “Kubism: Disassembling and Reassembling K-Means Clustering for Mobile Heterogeneous Platforms”, The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[C3] Dongwon Yang, Jaebeom Jeon, Minseong Gil, **Junsu Kim**, Seondeok Kim, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, “SSFFT: Energy-Efficient Selective Scaling for Fast Fourier Transform in Embedded GPUs”, The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[J1] Minseong Gil, Jaebeom Jeon, **Junsu Kim**, Sangun Choi, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, “TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs”, The IEEE Embedded Systems Letters (ESL), 2024.

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh. “VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing”. The 53rd International Conference on Parallel Processing (ICPP), 2024

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung. “HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer”. The 39th IEEE International Conference on Computer Design (ICCD), 2021

Preprints (Project Names Only)

[P1] Co-author, “Hardware Supports for Enabling Arbitrary Numeric Format on GPUs”

Work Experience

d-Matrix, Santa Clara, CA, United States

AI Systems Intern (Remote)

Jun. 2026 - Sep. 2026

Mentor: Nithesh Kurella

The University of British Columbia, Vancouver, BC, Canada

Research Assistant at Systems and Architectures (STAR) Lab

Sep. 2025 - Current

Advisor: Prof. Prashant Nair

Korea University, Seoul, Korea

Research Assistant at Computer Architecture and System Software Lab (ComSys)

Sep. 2023 - Aug. 2025

Advisor: Prof. Yunho Oh

Korea Institute of Science and Technology, Seoul, Korea

Research Assistant at Korea Data Science Team (KDST)

May. 2022 - Aug. 2023

Supervisor: Dr. Suhyun Kim

Pennsylvania State University, State College, PA, United States

Research Assistant

Sep. 2021 - Mar. 2022

Advisor: Prof. Swaroop Ghosh

Hanyang University, Seoul, Korea

Research Assistant at Embedded System on Chip Laboratory (ESOC Lab)

Research Assistant at Computer Architecture and System SW Lab (CASS Lab)

Dec. 2019 - Mar. 2020, Aug. 2020 - Nov. 2020

Advisor: Prof. Ki-Seok Chung

Advisor: Prof. Yongjun Park

Teaching Experience

The University of British Columbia, Vancouver, BC, Canada

Teaching Assistant for Computer Architecture (CPEN 411)

Fall 2025

Korea University, Seoul, Korea

Teaching Assistant for Computer Architecture

Spring 2024, Fall 2024

School for the Blind, Chuncheon, Korea

Assistant Teacher (Alternative Military Service)

Mar. 2017 - Feb. 2019

Honors and Awards

ISCA Student Travel Grant → \$1000 USD

May. 2026

Faculty of Applied Science Graduate Award, University of British Columbia (UBC) → \$7200 CAD

Nov. 2025

AAAI Student Travel Grant → \$1600 USD

Mar. 2025

Korea University Graduate School Scholarship → ~ \$5000 CAD/semester

Spring 2024, Fall 2024

Korea University Graduate School Scholarship for Outstanding New Student → ~ \$5000 CAD

Fall 2023

Hanyang University Scholarship → ~ \$5000 CAD/semester

2014 - 2019

Research Projects

Scalable Speculative Decoding with Dynamic Speculator Selection

Advisor: Prof. Prashant Nair, The University of British Columbia

Mar. 2026 - Current

◇ To be updated

Accuracy-preserving Tiered Key-Value Cache Management for Large Language Model

Advisor: Prof. Prashant Nair, The University of British Columbia

Sep. 2025 - Mar. 2026

- ◇ Proposed a tiered KV cache management framework that leverages a full KV cache stored in host DRAM and a compressed KV cache residing in GPU HBM
- ◇ Proposed reforwarding that rejects a new output token generated by the compressed KV cache and generates a new one with full KV cache depending on output token confidence
- ◇ Proposed KV cache recomputation that recomputes KV cache generated by the compressed KV cache during reforwarding
- ◇ Reduced the KV cache size by 60% while achieving an average speedup of 65.4% on LongWriter (Long-generation) and Longbench-v1 (Long-context), with less than 1% accuracy loss.
- ◇ Contributions: first-author, motivation study, idea, implementation

Solving Non-selection Problem for Probabilistic Rowhammer Defence [ISCA'26]

Advisor: Prof. Prashant Nair, The University of British Columbia

Nov. 2025 - Mar. 2026

- ◇ Identified the non-selection problem in fixed-rate probabilistic RowHammer defenses, where aggressively hammered rows can repeatedly escape random sampling at low RowHammer thresholds
- ◇ Proposed PrISM, a probabilistic RowHammer mitigation that triggers additional mitigations only under persistent row accesses
- ◇ Designed low overhead in-DRAM structures: Sampled Slot Queue, Sampled History Queue, and Pending Mitigation Queue
- ◇ Achieved average slowdown of 0.2% at a threshold of 500 and 1.5% at a threshold of 250
- ◇ Contributions: co-first author, related-work implementation, experiment, paper write-up

Tensor Migration Scheduling for GPU Unified Storage Architecture [CAL'25]

Advisor: Prof. Yunho Oh, Korea University

Feb. 2024 - Sep. 2024

- ◇ Analyzed page faults due to memory oversubscription stalled AI workloads when expanding GPU memory with SSD using UVM
- ◇ Proposed a tensor migration scheduling algorithm considering GPU memory oversubscription for GPU unified storage architecture
- ◇ Achieved the averaged speedup by 12.9% compared to G10, which was presented at MICRO 2023
- ◇ Contributions: 1st author, motivation study, idea, implementation, experiment, paper write-up

Hardware Supports for Enabling Arbitrary Numeric Format on GPUs

Advisor: Prof. Yunho Oh, Korea University

May. 2024 - Nov. 2024

- ◊ Observed GPU supports a limited set of numeric formats, wasting register files when processing arbitrary numeric formats
- ◊ Employed bitslice representation, which transposes the data elements, packing arbitrary numeric formats without register wastage.
- ◊ Proposed Bitslice Vector multiplier and adder, constructing a tree structure to replace a multiplication-adder tree in a Tensor core
- ◊ Contributions: co-author, motivation study, idea, implementation, paper write-up

A Behavioral Analysis of CXL Memory Systems

Advisor: Prof. Yunho Oh, Korea University

Sep. 2023 - Sep. 2024

Collaborator: SK hynix

- ◊ Observed the behavior of a real CXL-based system on datacenter and AI workloads in the CXL-based platform
- ◊ Analyzed how the different promotion and demotion methods for CXL devices affected the performance of the workloads
- ◊ Presented performance modeling for datacenter workloads using different system factors (e.g., memory bandwidth, memory latency)
- ◊ Contributions: co-author, experiment, analysis, paper write-up

Accelerating Yinyang K-Means on Heterogenous Platform [LCTES'25]

Advisor: Prof. Yunho Oh, Korea University

Jun. 2024 - Sep. 2024

- ◊ Analyzed that CPU was underutilized while executing Yinyang K-Means clustering on embedded GPUs
- ◊ Observed warp divergence caused by checking boundary conditions for skip clustering degraded performance
- ◊ Developed a software technique to cooperate with CPUs to enhance performance on resource-constrained embedded GPUs
- ◊ Proposed an adaptive reordering for the condition check to balance the warps
- ◊ Contributions: co-author, idea, implementation, paper write-up

TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs [ESL'24]

Advisor: Prof. Yunho Oh, Korea University

Mar. 2024 - Aug. 2024

- ◊ Observed fused kernels for multi-tenant AI workloads relied on the sub-optimal thread configuration
- ◊ Presented modeling to find the optimal thread configuration for fused AI kernels that balanced the warp-level computation
- ◊ Proposed a runtime system that dynamically fused AI kernels with the modeling
- ◊ Contributions: co-author, idea, paper write-up

VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing [ICPP'24]

Advisor: Prof. Yunho Oh, Korea University

Sep. 2023 - May. 2024

- ◊ Observed under-utilization of floating CUDA cores or Tensor cores when processing integer-quantized AI workloads
- ◊ Proposed a software technique for simultaneous computation on all heterogeneous cores on GPU to support arbitrary integer formats
- ◊ Proposed a software-based packing policy to support simultaneous processing of packed integers
- ◊ Contributions: co-author, motivation study, idea, implementation, paper write-up

Salient Frequency-aware Exemplar Compression for Online Continual Learning [AAAI'25]

Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology

Jan. 2023 - Nov. 2023

- ◊ Observed exemplar compression methods occupied limited GPU resources during online continual learning
- ◊ Proposed a computationally efficient compression algorithm using salient frequency
- ◊ Proposed a buffer management scheme to alleviate harmful effects from the compression artifacts remaining in the buffer
- ◊ Contributions: 1st author, motivation study, idea, implementation, paper write-up

Skills

C/C++, Python, Tensorflow, Pytorch, Git, Verilog, Shell script